

For each term from the lesson: write the definition AND give a numerical example that is DIFFERENT from the one used in the lesson.

1. The lesson worked through factor trees for 60 and 84. Now draw factor trees for these numbers and write the prime factorisation using index notation:

2. Use the Venn diagram method from the lesson to find HCF and LCM. Show prime factorisations and full working:

3. The lesson gives three real-world applications. For each problem below, state whether you need HCF or LCM, explain WHY, and calculate the answer:

4. The lesson says 1 is NOT a prime number because "it has only one factor: itself." Explain precisely why a prime number must have EXACTLY two factors. Why would including 1 as a prime cause problems in mathematics?

5. A student says: "To find the LCM of 12 and 18, I just multiply them: $12 \times 18 = 216$." Using the Venn diagram method from the lesson, show that the LCM is actually 36. Then explain what mistake the student made and why multiplying the two numbers together does NOT always give the LCM.

6. The lesson describes prime factorisation as "breaking a number into its atomic building blocks." Explain this analogy. In what way are prime numbers like atoms? Use the example of $60 = 2^2 \times 3 \times 5$ from the lesson in your explanation.
